

SIH Problem 26143: Oil Spill Detection & Vessel Attribution

Presentation Team Brief

The Problem (2 Slides)

What: Oil spills at sea go unattributed. We don't know who did it.

Why it matters:

Thousands of tonnes spilled yearly

Most polluters escape accountability

Current investigation takes weeks to months

Manual satellite analysis + manual AIS correlation = slow, expensive

Current gap:

Satellite imagery exists but isn't connected to ship tracking data

No automated system links detection to attribution

Our Solution (2 Slides)

The idea: Automated pipeline that detects oil spills from satellite → traces them backward to find origin point → matches that point with ship locations to identify the culprit.

Three stages:

Detection: AI model identifies oil slick from satellite imagery

Input: Sentinel-1 satellite data (free, public)

Output: Spill location, size, intensity

Time: ~2 minutes

Backtracking: Ocean physics simulates the spill backward in time

Input: Wind data, ocean current data (free, public)

Output: Where the spill started (origin point + time)

Time: ~30 minutes

Attribution: Match origin point with ship tracking data

Input: AIS vessel positions (free, public)

Output: Ranked list of suspect ships with confidence scores

Time: ~5 minutes

Total time: Under 1 hour (vs. current 40-80 days)

Why This Works

All proven technology:

ESA already detects oil spills with SAR imagery

Oceanographers use drift models for decades

Companies like MarineTraffic already track ships via AIS

We're just connecting them automatically

All free/public data:

Satellite imagery: Zenodo dataset + Sentinel-1

Ocean data: NOAA forecasts

Ship data: Marine Cadastre AIS database

Expected accuracy: 75-80% detection, top-3 suspect ranking 80% accurate

Technical Approach (1-2 Slides)

Detection model: U-Net (semantic segmentation)

Trained on ~70 labeled satellite images

Detects oil boundaries in satellite photos

Drift model: Lagrangian particle advection

Standard oceanography method

Runs ocean currents + wind backward in time

Attribution scoring: Bayesian ranking

Scores vessels by: proximity, trajectory, speed anomalies, vessel type

Explainable (judges can understand why rank #1 is ranked #1)

Tech stack: PyTorch, Python, PostgreSQL, Streamlit (all standard, open-source)

Impact & Benefits (1 Slide)

For government:

Reduces investigation time from weeks to hours

Automates correlation between satellite and AIS data

Enables prosecution with evidence

For environment:

Faster response to spills

Deters future incidents (polluters know they'll be caught)

Scale: Works globally (same algorithm, all oceans)

Key Numbers to Mention

3-4 million tonnes of oil spilled annually

40-80 days current investigation time

<1 hour our system's time

80% reduction in investigation burden

₹50+ lakhs current cost per investigation

75-80% expected detection accuracy

80% top-3 suspect ranking accuracy

What You Need to Know for Q&A

Why satellite + ocean data + AIS?

Only way to connect detection to attribution

Each data stream exists but is never combined

What if weather blocks satellite?

SAR (radar) works through clouds

And vessel tracking via AIS still works regardless

What if ships turn off AIS?

SAR still detects the spill

We can use port movements + historical patterns to narrow suspects

Confidence drops but tracking still possible

Timeline to build?

MVP: 8-12 weeks

Full system: 3-4 months

Who will actually use this?

NTRO (maritime surveillance)

Coast Guard (spill response)

International maritime authorities

Slide Structure (Official SIH Format - 6 slides max)

Slide 1: Title + problem ID + team info

Slide 2: Problem statement

What's happening: Oil spills unattributed

Why it matters: Scale + impact

What's missing: Connected data system

Slide 3: Proposed solution

Flowchart: Detect → Trace → Identify

Simple visual showing 3 stages

Output: Ranked suspect list

Slide 4: Technical approach

Architecture diagram (visual, not text-heavy)

Key technologies used + why

Data sources we're using

Slide 5: Impact & benefits

Time/cost reduction numbers

Who benefits (govt, environment, accountability)

Scalability

Slide 6: Research & references

3-5 key papers on oil detection / drift modeling / vessel tracking

Data sources (Zenodo, NOAA, Marine Cadastre)

Similar projects proving feasibility (ESA, OceanParcels)

Delivery Notes

Opening (30 seconds): "We're solving problem 26143. Oil spills happen. We don't know who did it. Our system automatically connects satellite detection to vessel tracking to prove guilt. Faster, cheaper, accountable."

Tone:

Technical but not jargon-heavy

Confident (this is solvable)

Impact-focused (this matters)

Avoid:

Claiming 99% accuracy

Over-explaining technical details

Reading from slides

Making excuses about not having code

Timing: 7-9 minutes talking + 2-3 min Q&A

Simple Q&A Answers

Q: Is this realistic?

A: "Yes. ESA detects spills. OceanParcels models drift. MarineTraffic tracks ships. We're automating all three together."

Q: What if your model fails?

A: "We validate against past incidents. Uncertainty is built in. If detection is $\pm 20\text{km}$, we cast wider net for suspects."

Q: How will maritime authorities use this?

A: "Same data they use now. We just automate the analysis. They get results in hours instead of days."

Q: What about false positives?

A: "Precision over recall. Better to miss a tiny spill than falsely accuse a ship."

Presentation Day Checklist

- ☐ Slides follow official SIH template
- ☐ No spelling errors
- ☐ All 6 slides have clear visuals (not text walls)
- ☐ Font size readable (18pt minimum)
- ☐ Practiced delivery 5+ times
- ☐ Can answer the Q&A above without reading notes
- ☐ Assigned backup speaker for each slide
- ☐ Dressed professionally
- ☐ PDF exported (not PowerPoint)
- ☐ Submitted by deadline

That's it. You know what to say. Build the slides. Practice. Done.